GEMSTAR Thesis — Completed Work, Tasks 01–20

Research progress and evidence register | 02 November 2026 | Companion to the Task 21–50 future roadmap

This document records what has been implemented, examined or documented through Task 20. It is not a claim that the full thesis, three-sensor object fusion, or ARM–FPGA experiments are complete. Tasks 08–20 are informed by available task archives and reports; Tasks 01–07 are reconstructed from the project history and should be cross-checked against original files.

Executive status

Area	Progress through Task 20
Simulation and datasets	GEMSTAR, Unreal Engine, SUMO, OSI SensorView/GroundTruth recordings; parsing and reproducible analysis pipelines.
Sensors	Camera image decoding/detection, LiDAR point-cloud reconstruction and filtering, Radar reflections/range/geometry audits.
MATLAB	Used already for camera preprocessing comparison, sensor-quality weighting, timing and LiDAR/software policy evaluation.
Radar engineering	Identified original hard-coded 10 m range and verified corrected configurable 50 m operation.
Research rigor	Timestamp audits, scenario inventories, source-code review and documented unresolved geometric association.
Not yet verified	Complete camera–LiDAR–radar object fusion; real FPGA acceleration, board energy savings and live end-to-end benchmark.

Phase A — Environment and original data foundation (Tasks 01–07)

Task 01 — Development environment and repository

Work: Set up GEMSTAR repository, Unreal Engine project and dependencies, including simulator build/toolchain preparation.

Evidence / output: Unreal project and feature plugins available locally.

Qualification: Specific Task 01 completion evidence is not present in the available archive.

Task 02 — GEMSTAR simulation integration

Work: Configured initial simulator execution and essential features, including SUMO traffic and camera/LiDAR/radar modules.

Evidence / output: GEMSTAR startup logs showed active sensor and SUMO features.

Qualification: This is a reconstructed high-level summary, not a line-by-line audit of Task 02.

Task 03 — Sensor and logging configuration

Work: Inspected DefaultGemstar.ini, sensor mounts, logger input routing and output directories.

Evidence / output: Identified camera/LiDAR/radar feature configuration and generated OSI log structure.

Qualification: Not every sensor was populated in every recording.

Task 04 — First OSI recordings and identification

Work: Generated/identified recorded SensorView and GroundTruth files and vehicle mappings.

Evidence / output: OSI files and sample GT vehicle mapping used in later tasks.

Qualification: Early recordings had missing camera/LiDAR SensorView payloads.

Task 05 — Initial OSI sensor parser

Work: Used Python osi3/protobuf tools to count frames and inspect sensor payload presence.

Evidence / output: One 542-frame file had 542 radar payloads but zero camera and LiDAR payload frames.

Qualification: A positive radar payload count does not prove target detection.

Task 06 — Data extraction and diagnostic checks

Work: Built first reusable parsing and sensor availability/health workflow.

Evidence / output: Preliminary frame-level metrics, extracted measurements and script outputs.

Qualification: Exact Task 06 subtasks should be verified against older local archive.

Task 07 — Scenario preparation and first comparisons

Work: Prepared clear/fog recording references and compared sensor behaviour before correction.

Evidence / output: Early scenario labels, configs and experimental scripts underpin Tasks 08–10.

Qualification: Recording conditions were not always paired or identical.

Phase B — Sensor behaviour and reliability (Tasks 08–10)

Task 08 — Clear/fog sensor experiments and LiDAR weather fix

Work: Compared camera brightness/contrast, LiDAR range/return rates, radar outputs and trajectories; reviewed weather data source in C++.

Evidence / output: LiDAR code was corrected to use current weather state. Pre/post-fix positive return rate 45.082% vs 18.966%, on different runs. Post-fix maximum positive range 6.927 m. Camera fog sample was brighter with lower contrast.

Qualification: Comparisons were not controlled paired runs; the simple radar did not show verified weather attenuation.

Task 09 — Sensor reliability/quality data exploration

Work: Prepared diagnostic datasets and MATLAB/Python measurements for quality-based weighting.

Evidence / output: Archived Task 09 scripts/data; detailed completion metrics not verified here.

Qualification: Quality proxies are not calibrated obstacle-detection probabilities.

Task 10 — Synchronized quality scores and fault injection

Work: Built 47 aligned samples for MOR100 and simulated camera, LiDAR and radar metric degradation; normalized quality-based sensor weights.

Evidence / output: Mean example quality scores: camera 0.782, LiDAR 0.861, radar 0.933; weighting reacted to injected faults and MATLAB normalization passed.

Qualification: Synthetic fault injection; not demonstrated improvement in object-level fusion accuracy.

Phase C — Cross-sensor geometry and evaluation (Tasks 11–14)

Task 11 — Three-sensor reconstruction and association prototype

Work: Parsed MOR100 recording, reconstructed candidate LiDAR/radar coordinates, inspected ground filtering and clustering, and tested camera YOLO associations.

Evidence / output: Archived a detailed report, visualizations, YOLO11 results and geometry audits around reference frame 150.

Qualification: Coordinates and associations remained provisional; no validated tri-sensor fusion.

Task 12 — Ground-truth proximity and camera projection

Work: Compared external GT vehicle with camera detection box and candidate LiDAR/radar returns.

Evidence / output: Camera projected centre was approximately 37.43 pixels from the candidate detection centre; examined 279/389 LiDAR points and 6/7 radar returns within a 2 m centre neighbourhood under alternative conventions.

Qualification: Nearby returns were not independently established as target-object reflections.

Task 13 — Geometric calibration and temporal association diagnostics

Work: Investigated common sensor frame, radar/LiDAR geometry, ground/actor ambiguity, visibility and multiframe association.

Evidence / output: Geometry/association analysis scripts, figures, MATLAB diagnostics and findings archived.

Qualification: Full extrinsic and object-surface identification not definitively established.

Task 14 — Extended radar, LiDAR and camera consistency checks

Work: Audited radar angular ray generation/source code, object proximity, LiDAR ground separation and scenario cross-sensor evidence.

Evidence / output: Sensor-specific geometry and coverage reports; MATLAB checks archived.

Qualification: Evidence classification is not verified perception accuracy.

Phase D — Dataset integrity and thesis positioning (Tasks 15–17)

Task 15 — Dataset inventory and integrity audit

Work: Indexed historical sensor recordings and file hashes, assessed scenario/config coverage.

Evidence / output: 49 OSI references, 38 distinct file hashes, 19 unique SensorView recordings; 10 had full three-sensor payload coverage, 9 did not.

Qualification: Complete payload coverage does not establish calibration or useful detections; some fog folders empty.

Task 16 — Research novelty and experiment design

Work: Compared literature on fusion, FPGA acceleration, CPU–FPGA scheduling, sensor uncertainty and mixed precision.

Evidence / output: Inventory of 20 papers, 5 priority competing works and provisional hypotheses/baselines.

Qualification: Novelty remains a hypothesis requiring rigorous literature verification and experimental evidence.

Task 17 — Timestamp and sensor coordinate audit

Work: Compared paired MOR100 466-frame SensorView and GroundTruth recordings, examined intervals, source mounts and coordinate assumptions.

Evidence / output: All 466 recorded timestamps matched; a change from approx. 30.49 Hz to 3.00 Hz appeared near frame 424.

Qualification: Timestamp labels alone do not confirm simultaneous sensing or end-to-end processing latency.

Phase E — Software perception and benchmarking (Tasks 18–19)

Task 18 — Camera pipeline and MATLAB/Python equivalence

Work: Decoded 640×480 RGB frames, implemented grayscale/Gaussian/Sobel in Python and MATLAB, evaluated YOLO11 Nano/Small on selected frames, profiled CPU behaviour.

Evidence / output: MATLAB-to-Python absolute differences for example frame: grayscale 0.000146, Gaussian 0.017559, Sobel 0.122012 on 0–255 scale; research reports archived.

Qualification: Small frame subset; limited camera ground truth and no FPGA camera accelerator.

Task 19 — LiDAR processing policy and early latency study

Work: Evaluated ground processing, voxel reduction, observability diagnostics and software scheduling across clear/fog recordings; studied quantization and HW partitioning.

Evidence / output: 48 frames selected, 47 quality-evaluated; modes FULL 22, VOXEL 25, SKIP 1; median software baseline/policy speed ratio about 1.398×; median monitor 0.15245 ms.

Qualification: Desktop MATLAB timings exclude I/O, transfers and fusion; no FPGA performance or energy data.

Phase F — Radar source correction and validation (Task 20)

Task 20 — Radar range correction, GT association research and final archive

Work: Inspected Unreal C++ ray traces, identified hard-coded range 10 m, patched configurable radar range flow and INI, rebuilt editor and generated 50 m recordings. Analyzed range distributions, startup artifacts, external GT position, FoV, ray geometry and approximate bounding-box intersections.

Evidence / output: Runtime range 50.000 m confirmed; first 100 frames: 5,696 candidate returns, 2,687 beyond 10 m, maximum 49.9283 m. Five eligible observations of one external car had no nearby 3 m reconstructed point; ray 44 showed returns at 41.2556 m rather than predicted vehicle box entry ~24.89–27.51 m. Final markdown/PDF summary and research audit completed according to conversation.

Qualification: Object-level detection accuracy not verified; target box assumption, pose timing and collision-hit identity unresolved. Not a hardware energy result.

Measured and reported outcomes — at a glance

Experiment	Reported result	Interpretation / caution
Task 08 fog LiDAR	45.082% → 18.966% positive returns	Pre- vs post-fix runs; scene differences remain.
Task 10 sensor weighting	47 aligned quality samples; dynamic weight redistribution	Response to synthetic quality faults; not detection gain.
Task 15 dataset coverage	10/19 recordings with full three-sensor payloads	Payload availability only.
Task 17 recorded timestamps	466/466 SV–GT matches	Message timestamps, not proof of sampling simultaneity.
Task 18 preprocessing	Python–MATLAB small numeric differences	Selected-frame equivalence only.
Task 19 processing policy	1.398× median software speed ratio	Desktop MATLAB partial pipeline, not FPGA.
Task 20 radar maximum	49.9283 m corrected recording	Validates recorded extension of range, not object identity.

Research readiness at the end of Task 20

Available foundations: recorded GEMSTAR sensor/GT data; reproducible OSI readers and CSV outputs; camera/LiDAR/radar processing experiments; MATLAB algorithms; fault-response quality weighting; geometry and timing audits; software scheduling prototypes; configurable radar C++ implementation.

Still required: full stable three-sensor datasets across varied maps and conditions; independent sensor extrinsic and object-hit validation; validated tracking and actual three-sensor fusion; rigorous comparable CPU baseline; ARM–FPGA accelerator and transfer integration; board-level power/energy evidence; adaptive scheduler with safety/quality constraints; end-to-end live demonstration.

Roadmap handover: where Task 21 begins

Task 21 starts with systematic MATLAB radar analysis and software benchmarking on the archived recordings. Later phases refine reusable multi-scenario data ingestion, verify camera/LiDAR/radar inputs and fusion, then move to ARM–FPGA hardware acceleration, measured energy/latency, adaptive scheduling and live GEMSTAR streaming. Consult the separate Task 21–50 roadmap for proposed numbered milestones.

Source basis and scope

Tasks 08–20: research files and summary reports available in the uploaded ALLLLL.zip archive, including Tasks 08, 10–20, and the conversation-confirmed Task 20.59–20.62 outcomes. Tasks 01–07 and selected Task 09 details: reconstructed from the project discussion and later artifacts; original task-specific archives were not available in this document build. This PDF deliberately avoids treating provisional comparisons, quality proxies or software timing as validated sensor-detection/FPGA-energy results.